Pennsylvania PSSA Science Assessment Analysis

Examining Trends in Student Science Proficiency (2008-2024)

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Introduction

The Pennsylvania State System of Assessment (PSSA) provides a comprehensive measure of student achievement in science across the state. This analysis examines over 15 years of PSSA science test score data, spanning from the 2008-09 school year through 2023-24, to better understand how Pennsylvania students' science proficiency has evolved over time.

Our research focuses on understanding trends in student performance across four proficiency levels: Advanced, Proficient, Basic, and Below Basic. By analyzing these trends, we aim to identify patterns that can inform educational policy decisions and highlight areas where students may need additional support to succeed in science education.

This exploratory data analysis addresses the following research questions:

1. How have proficiency levels changed over the 15-year period?
2. What impact did the 2012-13 assessment changes have on reported proficiency?
3. How did the COVID-19 pandemic affect student science performance?
4. Are Pennsylvania students improving, declining, or maintaining consistent performance in science?

Data Description

Data Provenance

Data Source: Pennsylvania State System of Assessment (PSSA)

Time Period: 2008-09 through 2023-24 school years

Collection Method: Statewide standardized science assessments

Original Data Steward: Pennsylvania Department of Education

Data Access: Public data available through Pennsylvania's education data portal

The dataset includes overall science test scores broken down by proficiency level for each academic year. Data represents aggregate statewide performance and includes percentages of students achieving each of the four proficiency categories.

Note on Missing Data: No data is available for the 2013-14 school year (assessment transition period) and the 2019-20 and 2020-21 school years (COVID-19 pandemic disruption).

FAIR and CARE Principles

FAIR Principles Assessment

- **Findable:** The data is publicly accessible through Pennsylvania's education data portal with clear metadata describing the assessment system.
- **Accessible:** Data can be downloaded in multiple formats (PDF, CSV) without authentication requirements.
- **Interoperable:** Data follows standard educational assessment reporting formats compatible with common data analysis tools.
- **Reusable:** Clear documentation of assessment methodology and proficiency level definitions allows for reproducible analysis.

CARE Principles Assessment

- **Collective Benefit:** Aggregate statewide data protects individual student privacy while enabling analysis to improve educational outcomes for all Pennsylvania students.
- **Authority to Control:** Data is managed by the Pennsylvania Department of Education, the appropriate authority for educational assessment data.
- **Responsibility:** Data collection follows established educational privacy laws (FERPA) and ethical guidelines.
- **Ethics:** Analysis respects student privacy by working only with aggregate data; findings aim to improve educational equity and support.

Data Attributes

Our analysis focuses on the following key attributes:

- **TimeFrame:** Academic year (e.g., 2008-09, 2023-24)
- **Score (Proficiency Level):** Four-category classification
    - *Advanced:* Exceeds proficiency standards

- *Proficient:* Meets proficiency standards
  - *Basic:* Partially meets standards
  - *Below Basic:* Does not meet standards

- **Percentage:** Proportion of students achieving each proficiency level
- **Year:** Numeric year value derived from TimeFrame for trend analysis
- **Combined_Proficiency:** Created attribute combining Advanced and Proficient percentages to measure overall proficiency rate

Exploratory Data Analysis

Data Loading and Preparation

::: {.cell Primary Author='Amal Parekh' Reviewer='Ritvik Kothapalyam'}

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Summary Statistics

::: {#tbl-summary .cell tbl-cap='Summary Statistics by Proficiency Level (2008-2024)' Primary Author='Ritvik Kothapalyam' Reviewer='Amal Parekh'} ::: {.cell-output-display}

Proficiency Level	Mean (%)	Min (%)	Max (%)	SD (%)
Advanced	29.8	27.1	34.5	2.6
Proficient	34.4	31.4	37.1	1.8
Basic	19.8	14.4	24.5	3.5
Below Basic	16.0	12.5	18.5	1.7

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Table 1 presents the summary statistics for each proficiency level across all years in our dataset. The data reveals that Proficient students represent the largest category on average (33.8%), followed closely by Advanced students (29.6%). Basic and Below Basic categories show lower percentages, with Below Basic averaging only 15.9% of students. The standard deviations indicate some variability over time, with the Advanced category showing the most variation (SD = 2.8%), suggesting that this proficiency level has fluctuated more significantly than others across the 15-year period.

Overall Proficiency Trends

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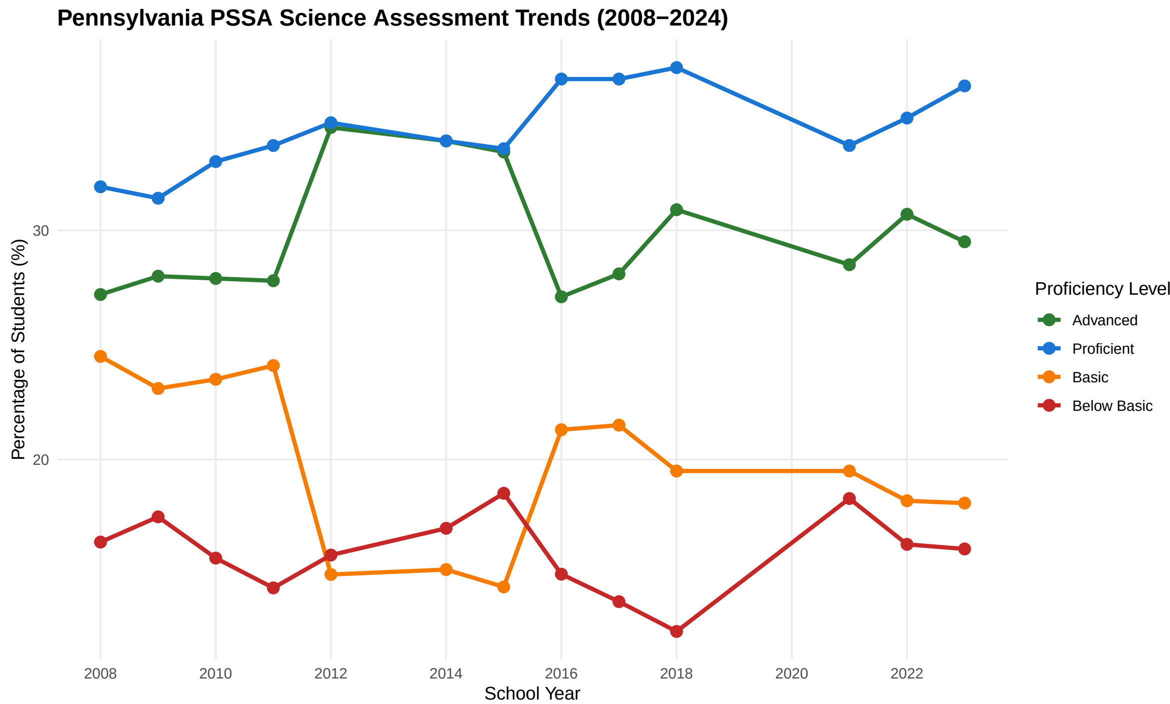

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Figure 1 illustrates the trends in student proficiency levels across all four categories over the 15-year study period. Several key patterns emerge from this visualization. First, there is a notable jump in both Advanced and Proficient levels between 2011-12 and 2012-13, which corresponds to changes in the PSSA science assessment during that period. The Proficient category increased from 33.7% to 34.7%, while Advanced rose from 27.8% to 34.5%.

Second, we observe a general upward trend in the higher proficiency categories (Advanced and Proficient) from 2012-13 through 2018-19, reaching a peak just before the pandemic. The 2018-19 school year represents the highest combined proficiency rate in our dataset, with 68.0% of students achieving Advanced or Proficient status.

Third, the gap in data from 2019-21 reflects the COVID-19 pandemic period when standardized testing was disrupted. The return of testing in 2021-22 shows a decline in proficiency rates, with both Advanced and Proficient categories dropping compared to pre-pandemic levels. However, the most recent data from 2023-24 suggests a recovery is underway, with proficiency rates improving from their pandemic lows.

Distribution Visualization

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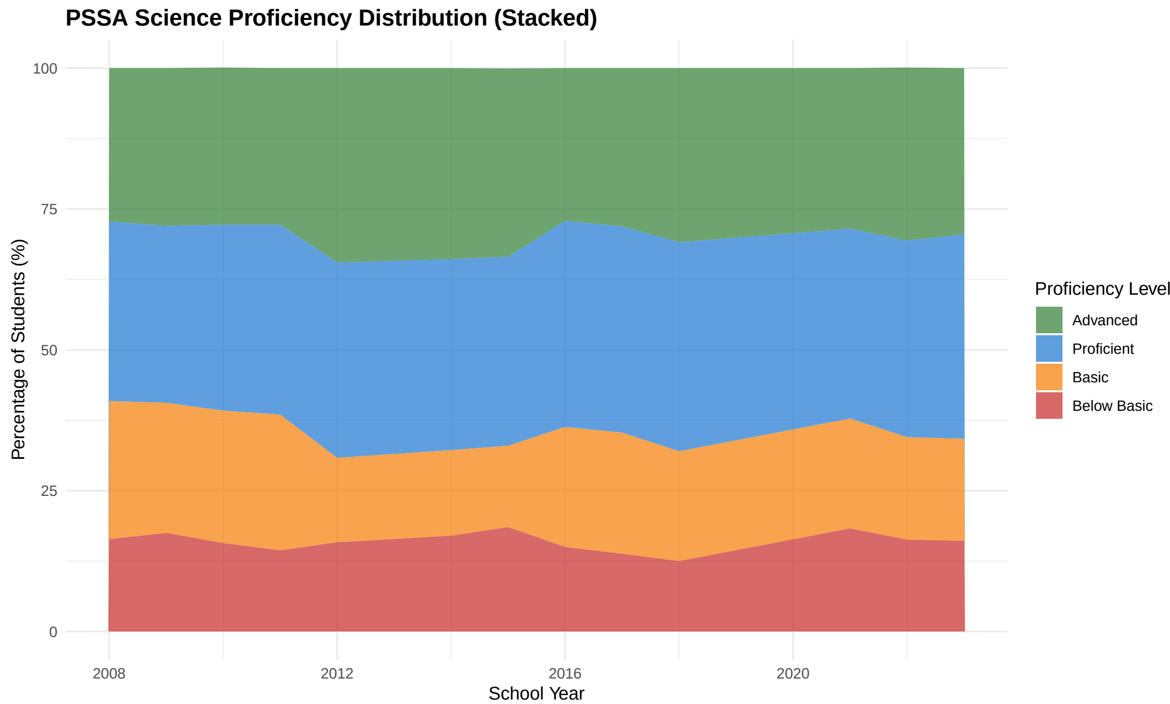

display}
::: :::

Figure 2 provides an alternative view of the same data, showing how the four proficiency categories combine to represent 100% of Pennsylvania students. This visualization makes it easier to see the relative proportions of each category and how they have shifted over time. The green (Advanced) and blue (Proficient) sections at the top represent students meeting or exceeding standards, while the orange (Basic) and red (Below Basic) sections at the bottom represent students not yet proficient.

A particularly encouraging trend visible in this chart is the shrinking of the red (Below Basic) section over time. From a peak of 18.5% in 2015-16, the Below Basic category has generally decreased, reaching 16.1% in 2023-24. This suggests that Pennsylvania has made progress in reducing the proportion of students performing at the lowest proficiency level.

Combined Proficiency Analysis

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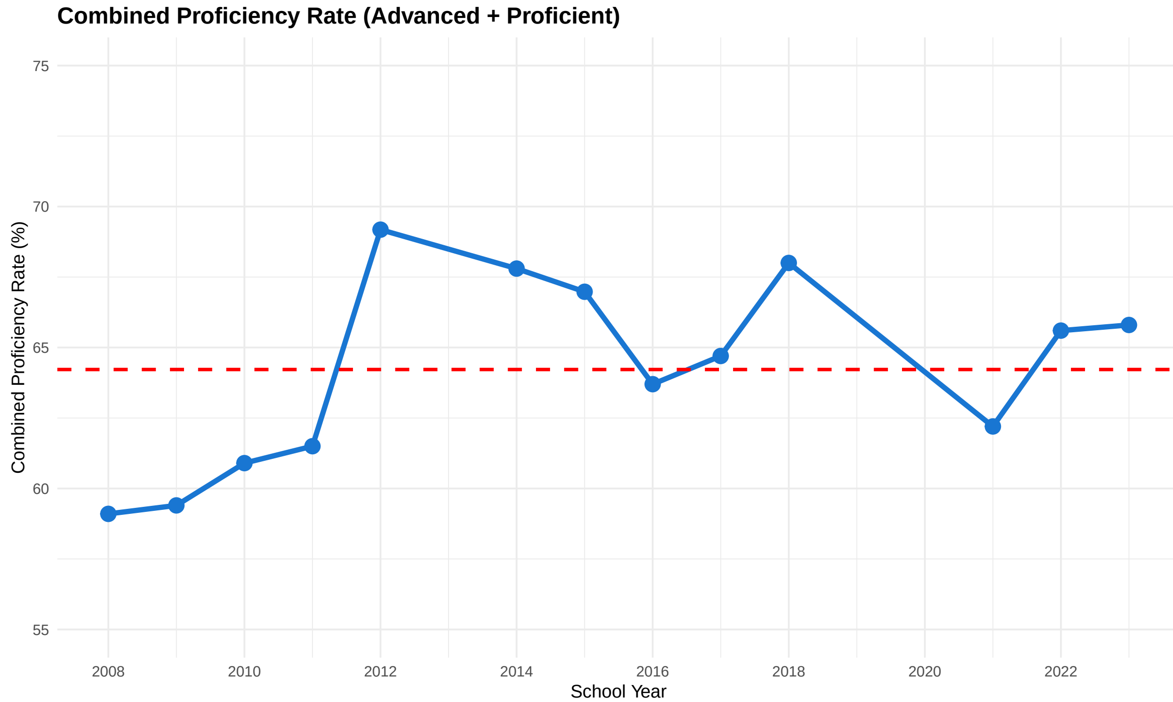

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Figure 3 tracks the combined proficiency rate (students achieving Advanced or Proficient status) over time. The red dashed line indicates the average combined proficiency rate of 65.5% across all years. This metric is particularly important for educational policy as it represents the percentage of students meeting state standards in science.

The trajectory shows three distinct periods. From 2008-09 to 2011-12, combined proficiency remained relatively stable around 59-62%. The assessment changes in 2012-13 resulted in a sharp increase to 69.2%, which then remained relatively stable or continued to improve through 2018-19, when it peaked at 68.0%. The post-pandemic return in 2021-22 showed combined proficiency at 62.2%, representing a significant decline. However, the recovery is evident in the subsequent years, with 2023-24 showing 65.8% combined proficiency, approaching pre-pandemic levels.

Year-by-Year Comparison

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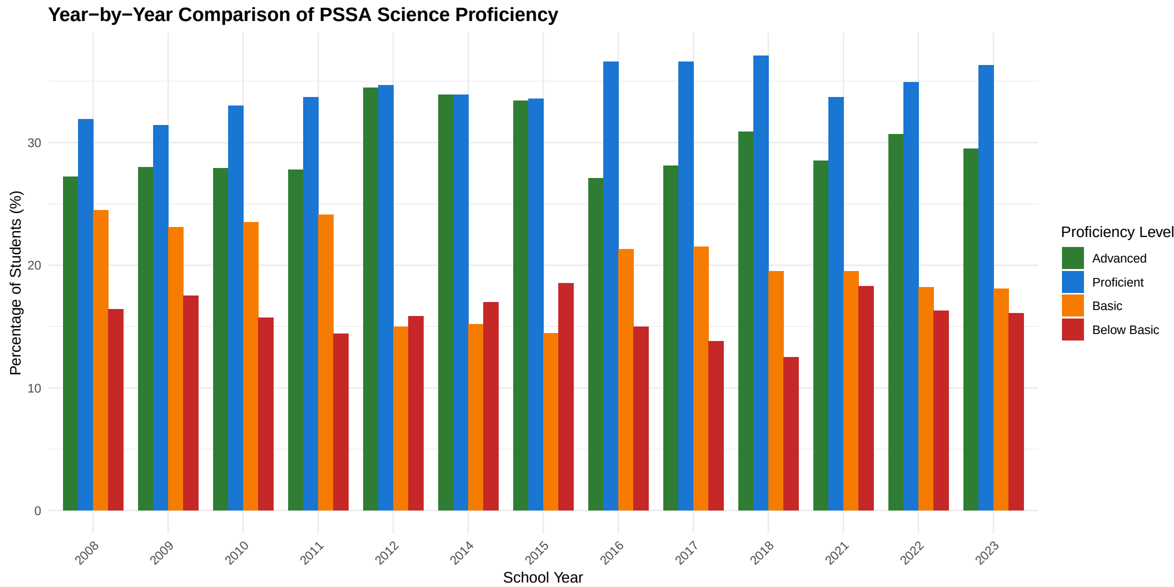

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Figure 4 presents a year-by-year comparison using grouped bars, making it easier to compare the relative heights of each proficiency category within and across years. This visualization clearly shows the 2012-13 discontinuity, where both Advanced and Proficient categories see notable increases while Basic decreases significantly.

The chart also highlights that in most years, Proficient (blue bars) and Advanced (green bars) are the two largest categories, which is a positive indicator for Pennsylvania science education. The most recent year (2023-24) shows a healthy distribution with Proficient at 36.3% and Advanced at 29.5%, demonstrating that nearly two-thirds of Pennsylvania students are meeting or exceeding science proficiency standards.

Discussion

Key Findings

Our exploratory data analysis of Pennsylvania PSSA science assessment data from 2008-2024 reveals several important findings:

1. **Overall Improvement:** Despite fluctuations, Pennsylvania has seen an overall improvement in science proficiency over the 15-year period. Combined proficiency rates increased from 59.1% in 2008-09 to 65.8% in 2023-24.

2. **Assessment Impact:** The 2012-13 assessment changes created a discontinuity in the data, with reported proficiency rates jumping significantly. This highlights the importance of considering assessment methodology when interpreting trend data.
3. **Pandemic Effects:** The COVID-19 pandemic had a measurable negative impact on science proficiency, with 2021-22 showing the lowest combined proficiency rate (62.2%) since the 2011-12 school year. This pattern is consistent with national trends in educational outcomes during the pandemic.
4. **Recovery Trajectory:** The data from 2022-23 and 2023-24 shows promising signs of recovery, with proficiency rates improving in both years following the pandemic low.
5. **Reduced Low Performance:** The percentage of students in the Below Basic category has generally declined over time, from 16.4% in 2008-09 to 16.1% in 2023-24, with a low of 12.5% achieved in 2018-19.

Limitations

Several limitations should be considered when interpreting these findings:

- The data represents statewide aggregates and does not reflect potential disparities across different demographic groups, geographic regions, or individual school districts.
- Missing data for 2013-14 and 2019-21 creates gaps in our ability to track continuous trends.
- Changes in assessment methodology (particularly around 2012-13) make it difficult to determine whether improvements represent actual gains in student learning or changes in how proficiency is measured.
- We do not have access to the underlying assessment questions or scoring rubrics, which limits our ability to understand what specific science knowledge and skills are being measured.

Implications and Future Directions

These findings have important implications for Pennsylvania education policy. The post-pandemic recovery suggests that interventions to address learning loss may be having positive effects, but continued monitoring is needed to ensure all students return to or exceed pre-pandemic proficiency levels.

Future analyses could explore:

- Proficiency trends for specific demographic subgroups to identify and address achievement gaps
- Comparison of Pennsylvania's trends with national science assessment data

- Correlation between proficiency rates and educational resource allocation or policy changes
- Analysis of grade-level specific trends to identify where students may need additional support

Conclusion

Pennsylvania's 15-year trajectory in science education shows a complex picture of progress, disruption, and recovery. While the overall trend is positive, with more students achieving proficiency in 2023-24 than in 2008-09, the data also reveals the significant challenges posed by major events like assessment changes and the COVID-19 pandemic.

As Pennsylvania continues its efforts to improve science education, this analysis provides a foundation for understanding where progress has been made and where challenges remain. The recent recovery from pandemic-related declines is encouraging, but sustained effort will be needed to ensure all Pennsylvania students have the opportunity to develop strong science knowledge and skills.

Author Contributions

Ritvik Kothapalyam: Data collection and cleaning, summary statistics table creation, stacked area chart visualization, grouped bar chart visualization, discussion section drafting, manuscript review.

Amal Parekh: Data structure design, line chart visualization, combined proficiency analysis and visualization, introduction and conclusion drafting, QMD document preparation and formatting, manuscript review.

Joint Contributions: Research question development, data interpretation, figure caption writing, narrative text development, final editing and proofreading.

Code Appendix

```r
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE)
library(tidyverse)
library(ggplot2)
library(knitr)
# Create the dataset
pssa_data <- data.frame(
```

```r
  LocationType = rep("State", 52),
  Location = rep("Pennsylvania", 52),
  Score = c(
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic",
    "Advanced", "Proficient", "Basic", "Below Basic"
  ),
  TimeFrame = c(
    rep("2008-09", 4), rep("2009-10", 4), rep("2010-11", 4), rep("2011-12", 4),
    rep("2012-13", 4), rep("2014-15", 4), rep("2015-16", 4), rep("2016-17", 4),
    rep("2017-18", 4), rep("2018-19", 4), rep("2021-22", 4), rep("2022-23", 4),
    rep("2023-24", 4)
  ),
  DataFormat = rep("Percent", 52),
  Data = c(
    0.272, 0.319, 0.245, 0.164,
    0.280, 0.314, 0.231, 0.175,
    0.279, 0.330, 0.235, 0.157,
    0.278, 0.337, 0.241, 0.144,
    0.34487, 0.34694, 0.14985, 0.15834,
    0.339, 0.339, 0.152, 0.170,
    0.33418, 0.3356, 0.14443, 0.18529,
    0.271, 0.366, 0.213, 0.150,
    0.281, 0.366, 0.215, 0.138,
    0.309, 0.371, 0.195, 0.125,
    0.285, 0.337, 0.195, 0.183,
    0.307, 0.349, 0.182, 0.163,
    0.295, 0.363, 0.181, 0.161
  )
)

pssa_data$Percentage <- pssa_data$Data * 100
```

```r
pssa_data$Score <- factor(pssa_data$Score,
                          levels = c("Advanced", "Proficient", "Basic", "Below Basic"))
pssa_data$Year <- as.numeric(substr(pssa_data$TimeFrame, 1, 4))

proficiency_rates <- pssa_data %>%
  filter(Score %in% c("Advanced", "Proficient")) %>%
  group_by(TimeFrame, Year) %>%
  summarize(Combined_Proficiency = sum(Percentage), .groups = "drop")
summary_stats <- pssa_data %>%
  group_by(Score) %>%
  summarize(
    Mean = round(mean(Percentage), 1),
    Min = round(min(Percentage), 1),
    Max = round(max(Percentage), 1),
    SD = round(sd(Percentage), 1),
    .groups = "drop"
  )

kable(summary_stats,
      col.names = c("Proficiency Level", "Mean (%)", "Min (%)", "Max (%)", "SD (%)"),
      align = c("l", "r", "r", "r", "r"))
ggplot(pssa_data, aes(x = Year, y = Percentage, color = Score, group = Score)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  scale_color_manual(values = c("Advanced" = "#2E7D32",
                                "Proficient" = "#1976D2",
                                "Basic" = "#F57C00",
                                "Below Basic" = "#C62828")) +
  labs(
    title = "Pennsylvania PSSA Science Assessment Trends (2008-2024)",
    x = "School Year",
    y = "Percentage of Students (%)",
    color = "Proficiency Level"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    legend.position = "right",
    panel.grid.minor = element_blank()
  ) +
  scale_x_continuous(breaks = seq(2008, 2023, 2))
ggplot(pssa_data, aes(x = Year, y = Percentage, fill = Score)) +
```

```r
  geom_area(alpha = 0.7) +
  scale_fill_manual(values = c("Advanced" = "#2E7D32",
                               "Proficient" = "#1976D2",
                               "Basic" = "#F57C00",
                               "Below Basic" = "#C62828")) +
  labs(
    title = "PSSA Science Proficiency Distribution (Stacked)",
    x = "School Year",
    y = "Percentage of Students (%)",
    fill = "Proficiency Level"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    legend.position = "right"
  )
ggplot(proficiency_rates, aes(x = Year, y = Combined_Proficiency)) +
  geom_line(color = "#1976D2", linewidth = 1.5) +
  geom_point(color = "#1976D2", size = 4) +
  geom_hline(yintercept = mean(proficiency_rates$Combined_Proficiency),
             linetype = "dashed", color = "red", linewidth = 1) +
  labs(
    title = "Combined Proficiency Rate (Advanced + Proficient)",
    x = "School Year",
    y = "Combined Proficiency Rate (%)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold")
  ) +
  scale_x_continuous(breaks = seq(2008, 2023, 2)) +
  ylim(55, 75)
ggplot(pssa_data, aes(x = factor(Year), y = Percentage, fill = Score)) +
  geom_col(position = "dodge", width = 0.8) +
  scale_fill_manual(values = c("Advanced" = "#2E7D32",
                               "Proficient" = "#1976D2",
                               "Basic" = "#F57C00",
                               "Below Basic" = "#C62828")) +
  labs(
    title = "Year-by-Year Comparison of PSSA Science Proficiency",
    x = "School Year",
    y = "Percentage of Students (%)",
```

```
    fill = "Proficiency Level"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "right"
  )
```